Chapter 4

$F-$distribution

In probability theory and statistics, the $F-$distribution, also known as Snedecor's F distribution or the Fisher–Snedecor distribution is a continuous probability distribution that arises frequently as the null distribution of a test statistic, most notably in the analysis of variance (ANOVA) and other $F-$tests.

4.1 F-Statistic

If X and Y are two independent χ^2-variates with n_1 and n_2 degrees of freedom respectively, then $F-$statistic is defined by

$$F = \frac{X/n_1}{Y/n_2}.$$

In other words, F is defined as the ratio of two independent χ^2-variates divided by their respective degrees of freedom and it follows Snedecor's $F-$distribution with n_1 and n_2 degrees of freedom, with probability function

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}, \ 0 \leq F \leq \infty.$$

Remarks: (i) The sampling distribution of $F-$statistic does not involve any population parameters and depends only on the degrees of freedom n_1 and n_2. (ii) A statistic F following Snedecor's $F-$distribution with n_1 and n_2 degrees of freedom will be denoted by $F \sim F\,(n_1, n_2)$.

4.2 Derivation of Snedecor's $F-$distribution

We have

$$F = \frac{x/n_1}{y/n_2} \quad \Rightarrow \frac{n_1}{n_2}F = \frac{x}{y}.$$

Therefore, $\frac{n_1}{n_2}F$ being the ratio of two independent χ^2-variates with n_1 and n_2 degrees of freedom respectively is a $\beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$ variate. Hence the probability function of F is given as

$$\begin{aligned} dP(F) &= \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{\left(\frac{n_1}{n_2}F\right)^{\frac{n_1}{2}-1}}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}\, d\left(\frac{n_1}{n_2}F\right) \\ &= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}\, dF,\ 0 \le F \le \infty. \end{aligned}$$

4.3 Moments of $F-$distribution

The rth raw moment of $F-$distribution can be obtain as follow

$$\begin{aligned} \mu_r' &= E\left(F^r\right) \\ &= \int_0^\infty F^r\ f(F)\ dF \\ &= \int_0^\infty F^r\ \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}\, dF \\ &= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \int_0^\infty \frac{F^{\frac{n_1}{2}+r-1}}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}\, dF. \end{aligned} \tag{4.1}$$

To evaluate the integral, set

$$\frac{n_1}{n_2}F = y,\ \text{so that } dF = \frac{n_2}{n_1}dy.$$

By using (4.1), further obtain as

$$
\begin{aligned}
\mu_r' &= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2},\frac{n_2}{2}\right)} \int_0^\infty \frac{\left(\frac{n_2}{n_1}y\right)^{\frac{n_1}{2}+r-1}}{(1+y)^{\frac{n_1+n_2}{2}}} \cdot \frac{n_2}{n_1}\, dy \\
&= \frac{\left(\frac{n_2}{n_1}\right)^{r}}{B\left(\frac{n_1}{2},\frac{n_2}{2}\right)} \int_0^\infty \frac{(y)^{\frac{n_1}{2}+r-1}}{(1+y)^{\frac{n_1+n_2}{2}}}\, dy \\
&= \frac{\left(\frac{n_2}{n_1}\right)^{r}}{B\left(\frac{n_1}{2},\frac{n_2}{2}\right)} \int_0^\infty \frac{(y)^{\frac{n_1}{2}+r-1}}{(1+y)^{\frac{n_1}{2}+r+\frac{n_2}{2}-r}}\, dy \\
&= \left(\frac{n_2}{n_1}\right)^{r} \cdot \frac{1}{B\left(\frac{n_1}{2},\frac{n_2}{2}\right)} \cdot B\left(\frac{n_1}{2}+r, \frac{n_2}{2}-r\right) \\
&= \left(\frac{n_2}{n_1}\right)^{r} \frac{\Gamma\left(\frac{n_1}{2}+r\right)\ \Gamma\left(\frac{n_2}{2}-r\right)}{\Gamma\left(\frac{n_1}{2}\right)\ \Gamma\left(\frac{n_2}{2}\right)}, \quad n_2 > 2r. \qquad (4.2)
\end{aligned}
$$

In particular

$$
\begin{aligned}
\mu_1' &= \left(\frac{n_2}{n_1}\right) \frac{\Gamma\left(\frac{n_1}{2}+1\right)\ \Gamma\left(\frac{n_2}{2}-1\right)}{\Gamma\left(\frac{n_1}{2}\right)\ \Gamma\left(\frac{n_2}{2}\right)} \\
&= \left(\frac{n_2}{n_1}\right) \frac{\frac{n_1}{2}\ \Gamma\left(\frac{n_1}{2}\right)\ \Gamma\left(\frac{n_2}{2}-1\right)}{\Gamma\left(\frac{n_1}{2}\right)\ \left(\frac{n_2}{2}-1\right)\ \Gamma\left(\frac{n_2}{2}-1\right)}, \qquad [\Gamma(n)=(n-1)\Gamma(n-1)] \\
&= \left(\frac{n_2}{n_1}\right) \frac{\frac{n_1}{2}}{\frac{n_2}{2}-1} \\
&= \frac{n_2}{n_2-2}, \quad n_2 > 2.
\end{aligned}
$$

Thus the mean of $F-$distribution is independent of n_1.

$$
\begin{aligned}
\mu_2' &= \left(\frac{n_2}{n_1}\right)^{2} \frac{\Gamma\left(\frac{n_1}{2}+2\right)\ \Gamma\left(\frac{n_2}{2}-2\right)}{\Gamma\left(\frac{n_1}{2}\right)\ \Gamma\left(\frac{n_2}{2}\right)} \\
&= \left(\frac{n_2}{n_1}\right)^{2} \frac{\left(\frac{n_1}{2}+1\right)\ \left(\frac{n_1}{2}\right)}{\left(\frac{n_2}{2}-1\right)\ \left(\frac{n_2}{2}-2\right)} \\
&= \frac{n_2^2\ (n_1+2)}{n_1\ (n_2-2)\ (n_2-4)}, \quad n_2 > 4.
\end{aligned}
$$

Therefore

$$
\begin{aligned}
\mu_2 &= \mu_2' - (\mu_1')^2 \\
&= \frac{n_2^2\ (n_1+2)}{n_1\ (n_2-2)\ (n_2-4)} - \left(\frac{n_2}{n_2-2}\right)^2
\end{aligned}
$$

$$= \frac{n_2^2\ (n_1+2)\ \ (n_2-2) - n_2^2\ n_1\ (n_2-4)}{n_1\ (n_2-2)^2\ (n_2-4)}$$
$$= \frac{n_2^2\left\{(n_1\ n_2 - 2n_1 + 2n_2 - 4) - (n_1\ n_2 - 4n_1)\right\}}{n_1\ (n_2-2)^2\ (n_2-4)}$$
$$= \frac{n_2^2(2n_2+2n_1-4)}{n_1\ (n_2-2)^2\ (n_2-4)}$$
$$= \frac{2n_2^2(n_2+n_1-2)}{n_1\ (n_2-2)^2\ (n_2-4)}, \quad n_2 > 4.$$

Similarly, on setting $r = 3$ and 4 in (4.2), we get μ_3' and μ_4' respectively, from which the central moments μ_3 and μ_4 can be obtained.

4.4 Mode of $F-$distribution

We have the probability density function of $F-$distribution as

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}$$
$$\Rightarrow \log\left(f(F)\right) = C + \left(\frac{n_1}{2}-1\right)\log F - \left(\frac{n_1+n_2}{2}\right)\ \log\left(1+\frac{n_1}{n_2}F\right),$$

where C is a constant independent of F.

$$\Rightarrow \log\left(f(F)\right) = C + \left(\frac{n_1}{2}-1\right)\log F - \left(\frac{n_1+n_2}{2}\right)\log\left(1+\frac{n_1}{n_2}F\right)$$
$$\Rightarrow \frac{f'(F)}{f(F)} = \left(\frac{n_1-2}{2}\right)\ \frac{1}{F} - \left(\frac{n_1+n_2}{2}\right)\ \frac{1}{\left(1+\frac{n_1}{n_2}F\right)}\ \frac{n_1}{n_2}$$
$$\Rightarrow 0 = \frac{n_1-2}{2F} - \frac{n_1(n_1+n_2)}{2(n_2+n_1F)}$$
$$\Rightarrow 0 = n_2(n_1-2) + n_1(n_1-2)F - n_1(n_1+n_2)F$$
$$\Rightarrow 0 = n_2(n_1-2) - n_1\left(n_2+2\right)F$$
$$\therefore F = \frac{n_2(n_1-2)}{n_1\left(n_2+2\right)}.$$

It can be easily verified that at this point $f''(F) < 0$.

Hence mode is $\frac{n_2(n_1-2)}{n_1(n_2+2)}$.

Example

Let X_1, X_2, X_3, X_4 and Y_1, Y_2, Y_3, Y_4, Y_5 be two random samples of size 4 and 5 respectively, from a standard normal population. Obtain the distribution and variance of the statistic $T = \left(\frac{5}{4}\right) \frac{X_1^2+X_2^2+X_3^2+X_4^2}{Y_1^2+Y_2^2+Y_3^2+Y_4^2+Y_5^2}$.

Solution: Since the population is standard normal, we get

$$X_1^2 + X_2^2 + X_3^2 + X_4^2 \sim \chi_4^2,$$

similarly,

$$Y_1^2 + Y_2^2 + Y_3^2 + Y_4^2 + Y_5^2 \sim \chi_5^2.$$

Thus

$$\begin{aligned} T &= \left(\frac{5}{4}\right) \frac{X_1^2 + X_2^2 + X_3^2 + X_4^2}{Y_1^2 + Y_2^2 + Y_3^2 + Y_4^2 + Y_5^2} \\ &= \frac{\frac{X_1^2+X_2^2+X_3^2+X_4^2}{4}}{\frac{Y_1^2+Y_2^2+Y_3^2+Y_4^2+Y_5^2}{5}} \\ &= \frac{\chi_4^2/4}{\chi_5^2/5} \\ &\sim F_{4,5}. \end{aligned}$$

Hence the statistic T follows $F-$ distribution with 4 and 5 degrees of freedom. Therefore we can obtain the variance as follow

$$\begin{aligned} Var(T) &= Var\left[F_{4,5}\right] \\ &= \frac{2(5)^2(5+4-2)}{4(5-2)^2(5-4)} \\ &= \frac{2(5)^2(7)}{4(3)^2(1)} \\ &= \frac{350}{36} \\ &= 9.72. \end{aligned}$$

Theorem 4.4.1. *Let $X \sim N(\mu_1, \sigma_1^2)$ and $X_1, X_2, \cdots, X_n$ be a random sample of size n from the population X. Also, let $Y \sim N(\mu_2, \sigma_2^2)$ and $Y_1, Y_2, \cdots, Y_m$ be a random sample of size m from the population Y. Then the statistic*

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n-1,\ m-1),$$

where S_1^2 and S_2^2 are the sample variances, respectively.

Proof. Since $X_i \sim N(\mu_1, \sigma_1^2)$, we get as follow

$$\frac{(n-1)S_1^2}{\sigma_1^2} \sim \chi^2_{(n-1)}.$$

Similarly, since $Y_i \sim N(\mu_2, \sigma_2^2)$, we get as follow

$$\frac{(m-1)S_2^2}{\sigma_2^2} \sim \chi^2_{(m-1)}.$$

Therefore

$$\frac{\frac{S_1^2}{\sigma_1^2}}{\frac{S_2^2}{\sigma_2^2}} = \frac{\frac{\chi^2_{(n-1)}}{(n-1)}}{\frac{\chi^2_{(m-1)}}{m-1)}} \sim F(n-1,\ m-1). \tag{4.3}$$

This completes the proof of the theorem. □

4.5 Relation Between F and t−distribution

We have the density function of F−distribution as

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}},\ 0 \le F \le \infty. \tag{4.4}$$

Take $n_1 = 1,\ n_2 = n$ and

$$F = t^2 \quad \Rightarrow \frac{dF}{dt} = 2t.$$

Thus the probability density function of F in (4.4) transforms to:

$$\begin{aligned} f(t) &= \frac{\left(\frac{1}{n}\right)^{\frac{1}{2}}}{B\left(\frac{1}{2},\frac{n}{2}\right)} \cdot \frac{(t^2)^{\frac{1}{2}-1}}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}\, 2t,\ 0 \le t^2 \le \infty \\ &= \frac{1}{\sqrt{n}\ B\left(\frac{1}{2},\frac{n}{2}\right)} \cdot \frac{1}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}\, dt,\ -\infty \le t \le \infty, \end{aligned}$$

the factor 2 vanishes since the total probability in the range $(-\infty, \infty)$ is 1. This is the density function of Student's $t-$distribution with n degrees of freedom.

Relation between t and F distributions: *If a statistic t follows Student's $t-$distribution with n degrees of freedom, then t^2 follows Snedecor's $F-$distribution with $(1, n)$ degrees of freedom. Symbolically, if $t \sim t_{(n)}$ then $t^2 \sim F_{(1,n)}$.*

4.6 Relation Between F and χ^2-distribution

In $F_{(n_1,n_2)}$ distribution if we let $n_2 \to \infty$, then $\chi^2 = n_1 F$ follows χ^2-distribution with n_1 degrees of freedom.

***Assignment: Prove the above statement.

4.7 Applicaitons of $F-$distribution

$F-$distribution has so many applications in statistical theory. For examples, testing the: (i) equality of two population variances; (ii) significance of an observed multiple correlation coefficient; (iii) significance of an observed sample correlation ratio; (iv) linearity of regression; (v) equality of several means and so on.